



Computer Science & IT



Data Structure & Programming

Tree

Lecture No. 07



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Recap of Previous Lecture



Topic

AVL tree

Topic

Rotation of AVL tree

Topic

Topic

Topic

Topics to be Covered



Topic

Counting of BST

Topic

Complexity

Topic

Minimum No. of Nodes in AVL tree of height h

Topic

Topic

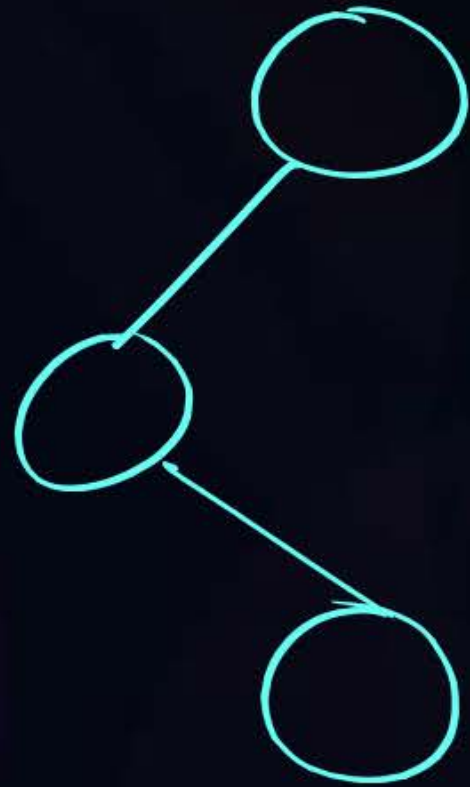


Enumeration of BST with n distinct Node

No. of unlabeled trees : $C_n = \frac{1}{n+1} 2^n C_n$

1 unlabeled tree, 3 distinct keys are given

11, 24, 26



No. of ways to label so that it becomes BST



Enumeration of BST with n distinct Node

No. of unlabeled trees : $C_n = \frac{1}{n+1} 2^n C_n$ 11, 24, 26



Each unlabeled can be Labeled in only 1 way So that it becomes a

BST : No. of BST with n node is $C_n = \frac{1}{n+1} 2^n C_n$

Question

Given 4 Distinct element {1,2,3,4}, how many BST will be of height 3. [8]

4 element and height 3 $\Rightarrow$ skewed



Enumeration of BST with n distinct Node



Number of BST with $\{1, 2, 3, \dots, n\}$ elements of height $n - 1$ is 2^{n-1}



Smallest AVL Tree of Height h

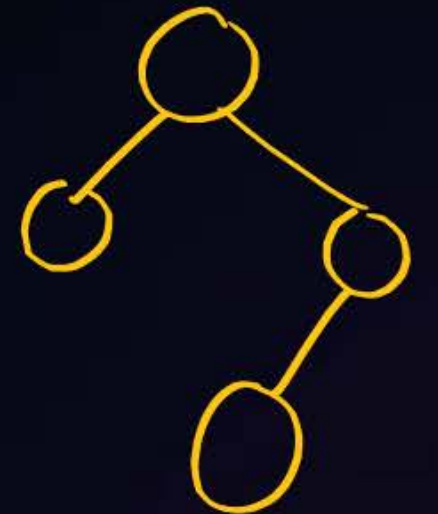
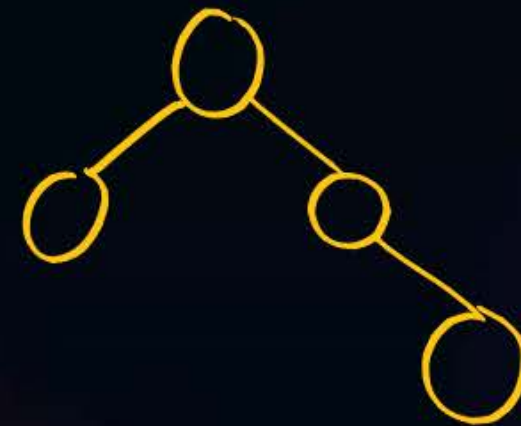
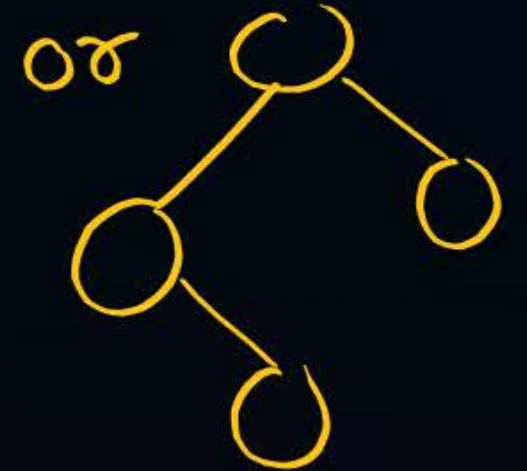


Max - FBT

AVL tree of Height 1 with Min Node :

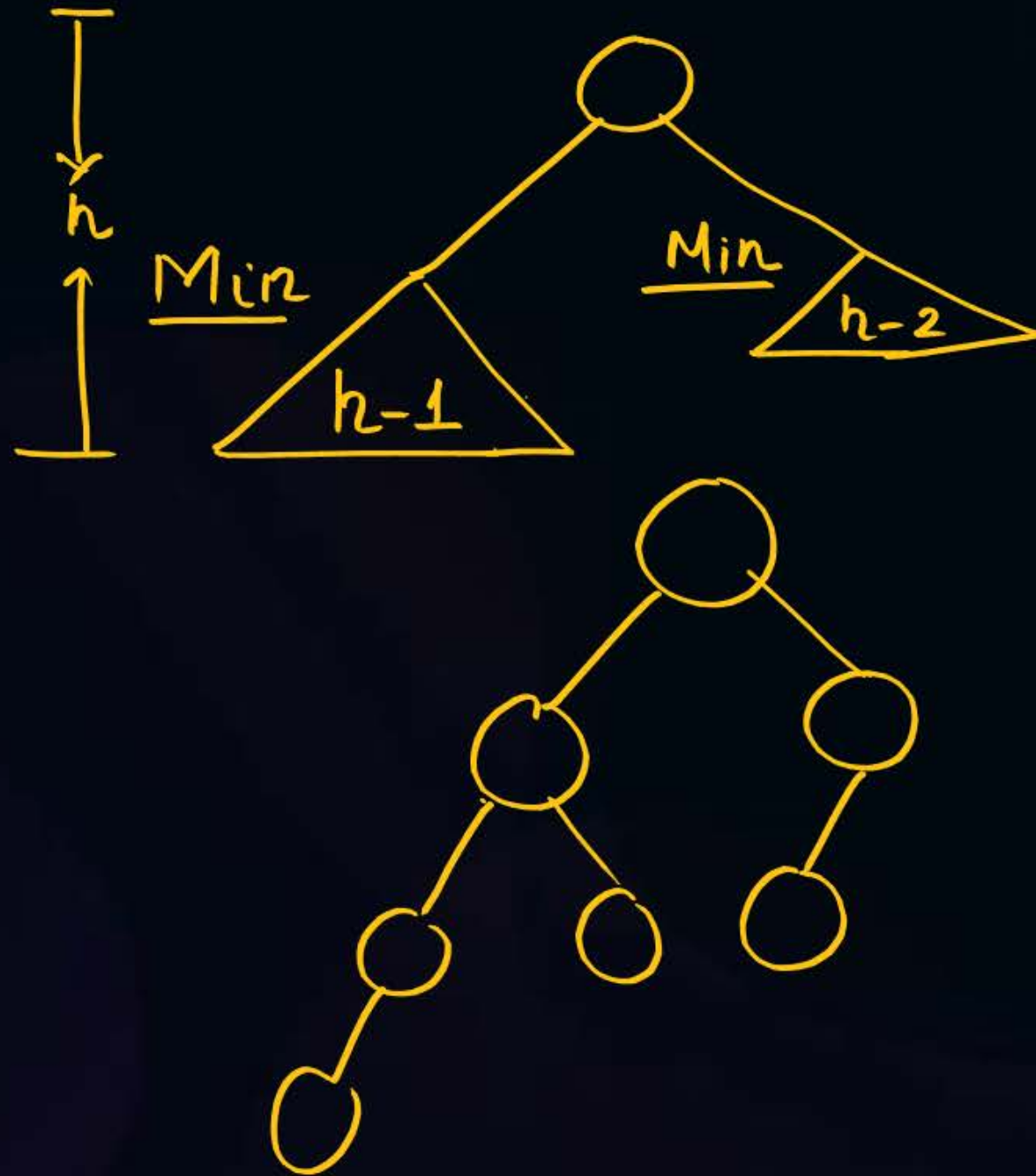


AVL tree of height 2 with min Node :

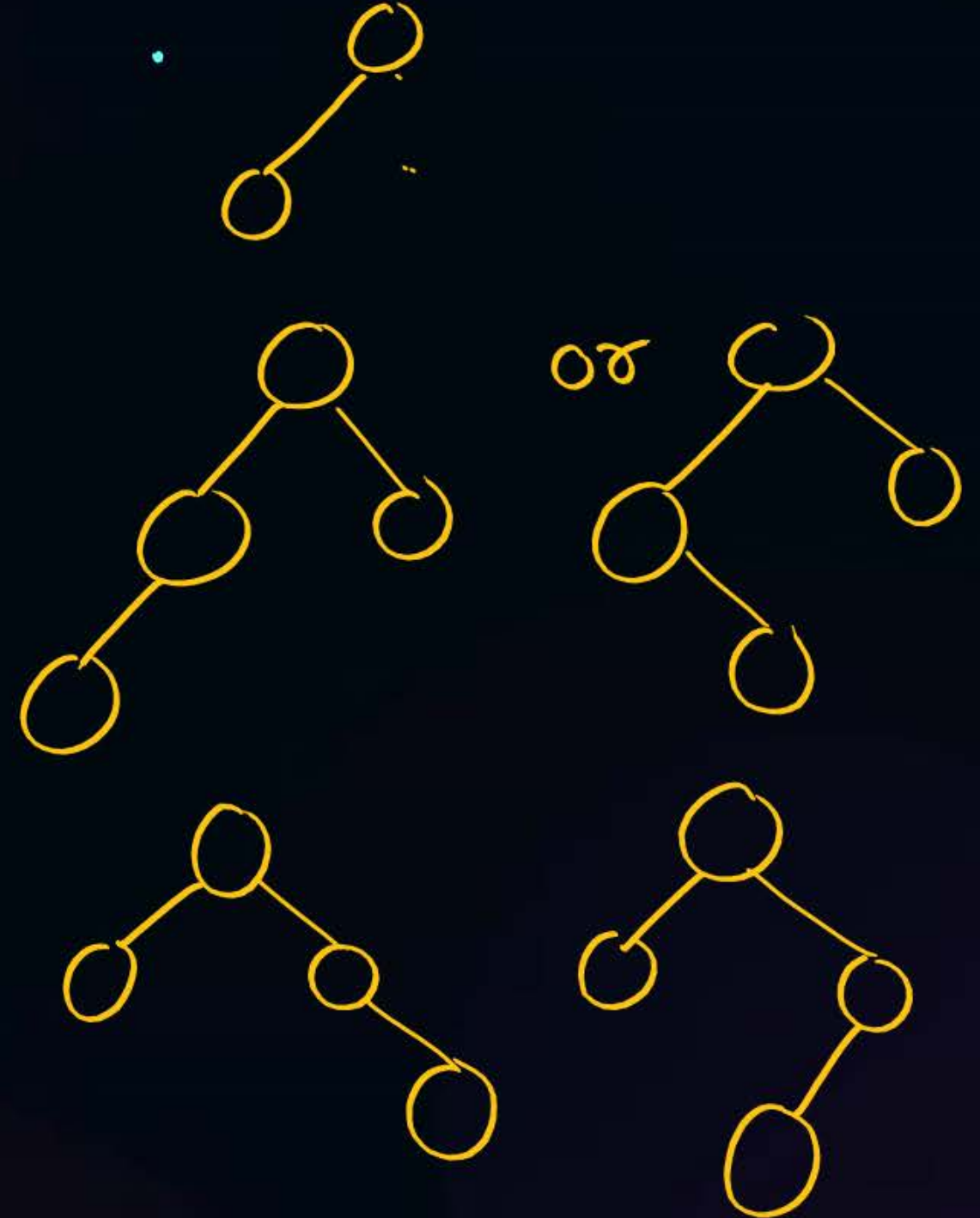




Smallest AVL Tree of Height h



Max - FBT





Smallest AVL Tree of Height h



$N(h)$ is minimum No. of nodes in height h

$$N(h) = N(h-1) + N(h-2) + 1$$

$$N(0) = 1 \quad \text{Base condition}$$

$$N(1) = 2$$

Min No. of node in height 8

$$N(2) = 4$$

$$N(7) = 54$$

$$N(3) = 7$$

$$N(8) = 88 \checkmark$$

$$N(4) = 12$$

$$N(5) = 20$$

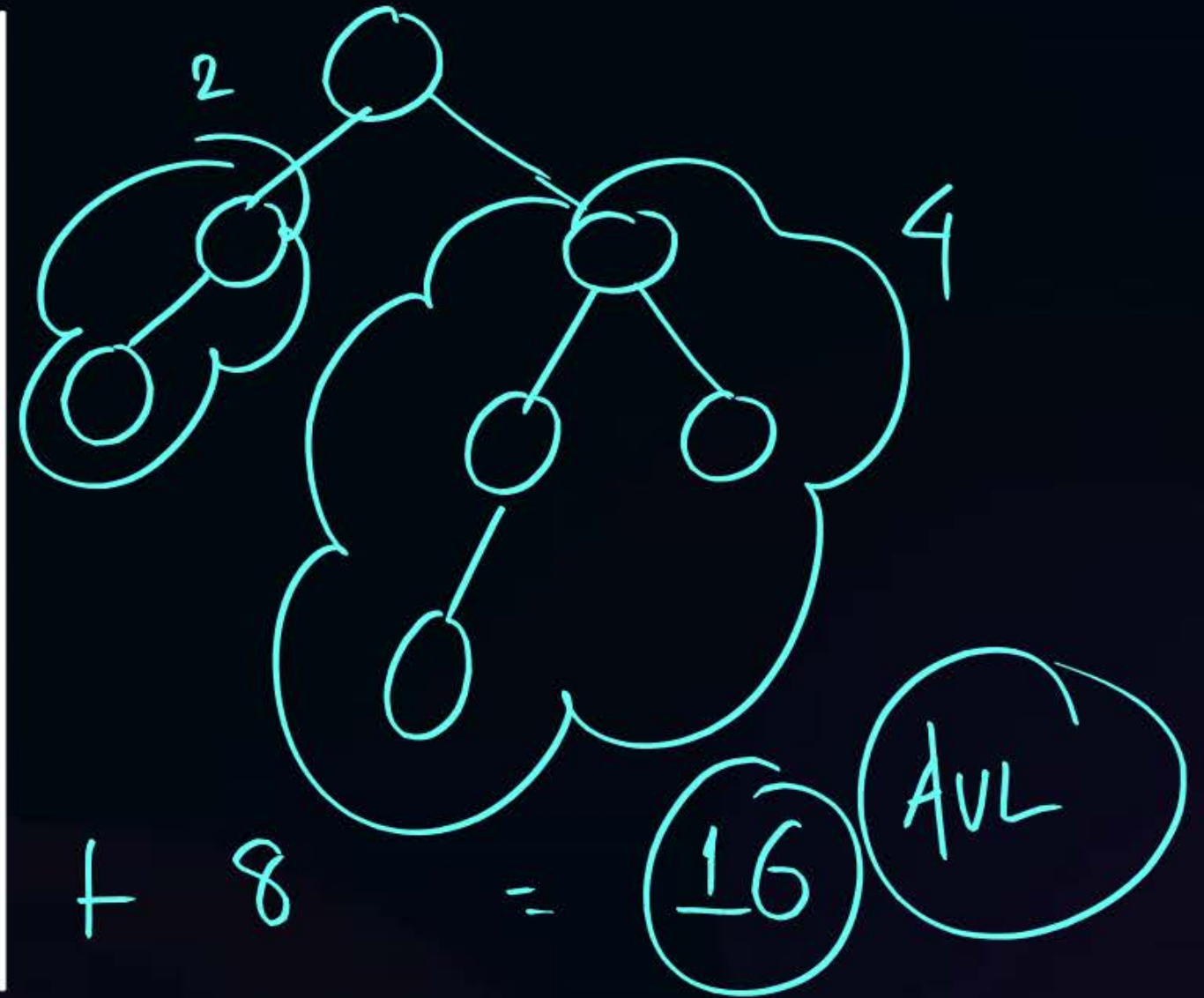
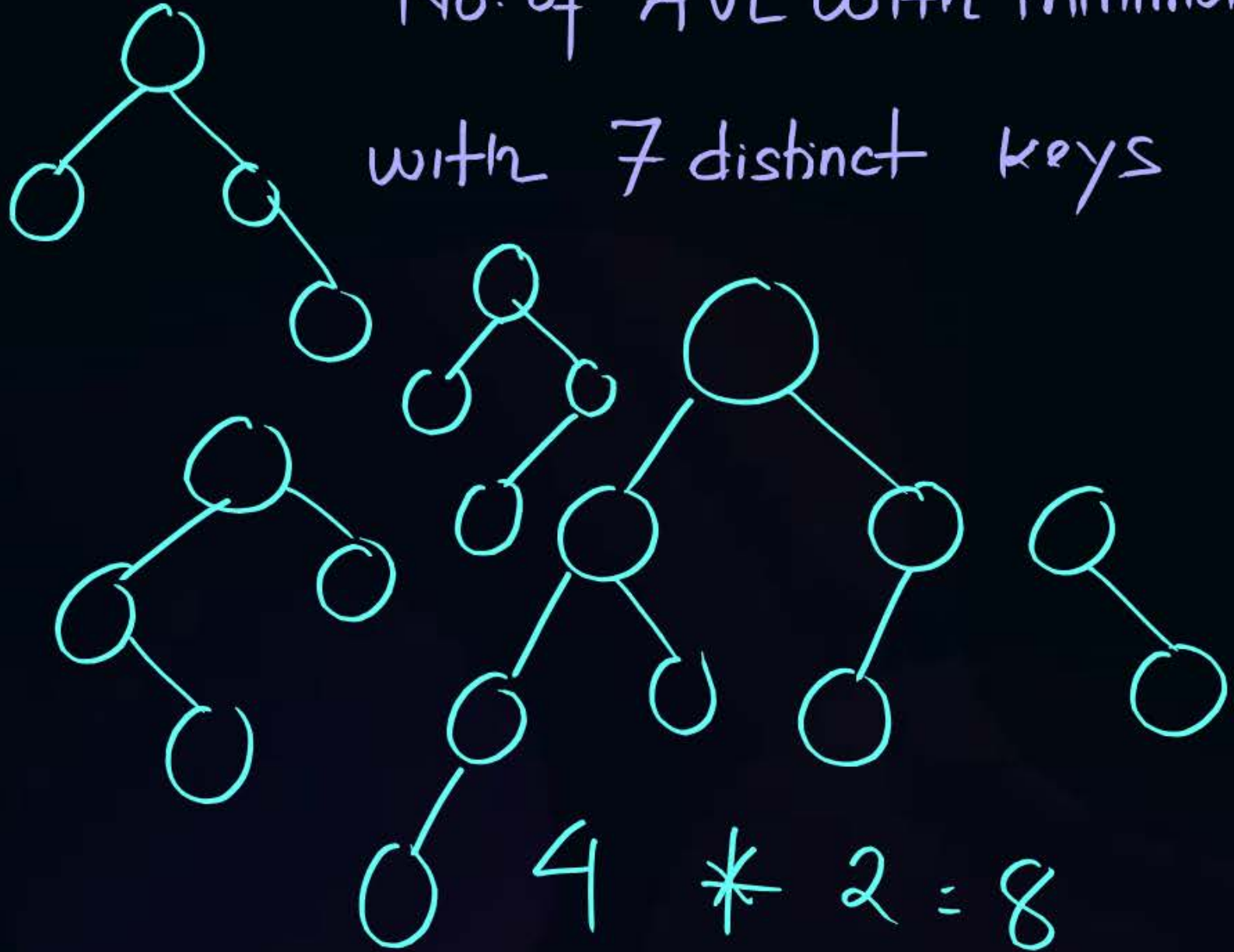
$$N(6) = 33$$



Smallest AVL Tree of Height h

No. of AVL with minimum No. of Node of height 3

with 7 distinct keys





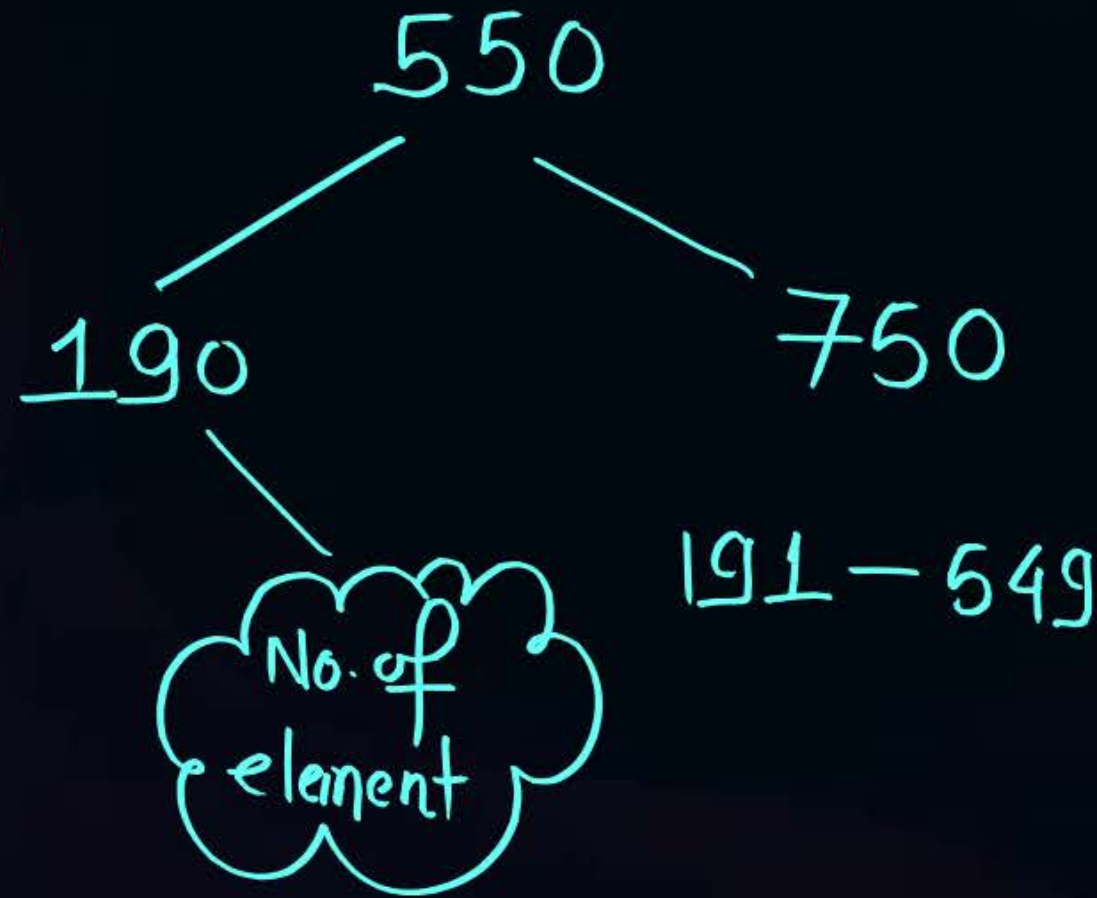
Topic : Question



$$5 - 10 \quad 10 - 5 + 1 \text{ (6)}$$

#Q A binary search is constructed with key values 1 to 1000 (each key is taken only once) if the root of the tree 550 and key value at left child of root is 190 and key value at right child of root is 750. Then number of elements in right subtree of left child of root is [359].

$$\text{UB} - \text{LB} + 1$$



$$549 - 191 + 1$$

$$\begin{array}{r} 550 \\ 191 \\ \hline 359 \end{array}$$

$$191 - 549$$

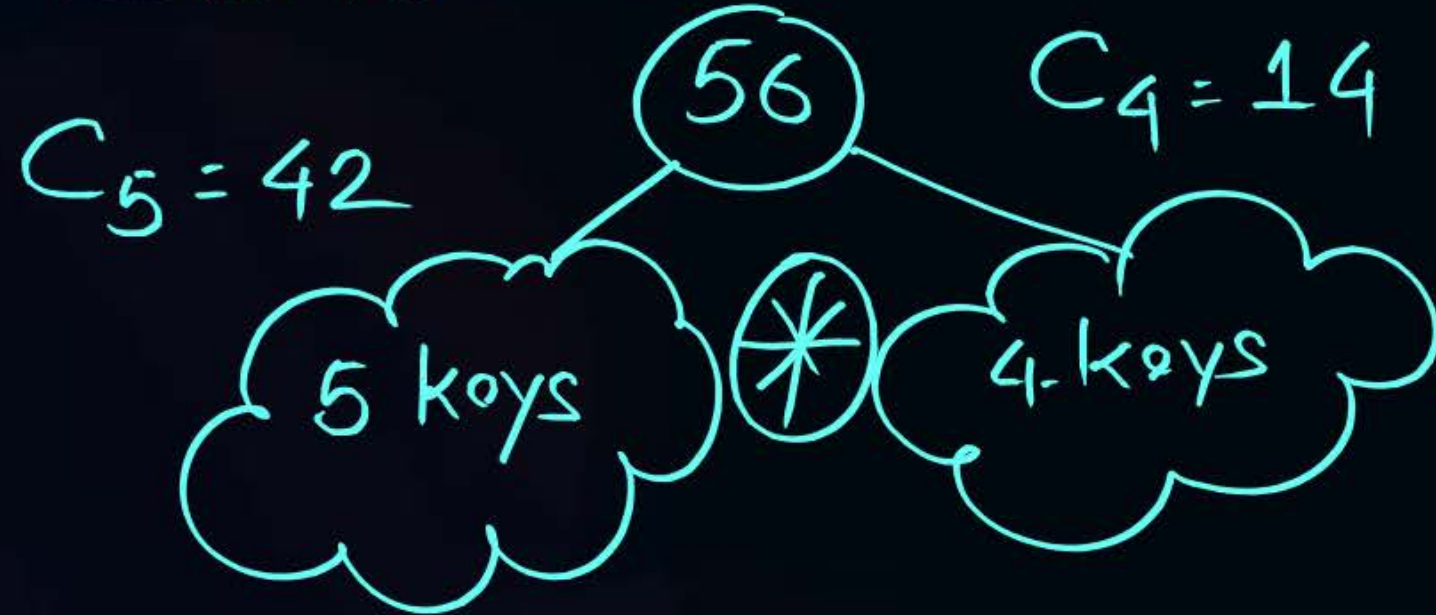


Topic : Question



#Q suppose BST is Constructed with 45, 23, 67, 21, 78, 98, 56, 43, 19, 87. If 56 is the root of the total number of BST possible with in order traversal as

19, 21, 23, 43, 45, 56, 67, 78, 87, 98 is 588



total No. of
BST possible
 42×14
588²

GATE 2004, 2-Marks



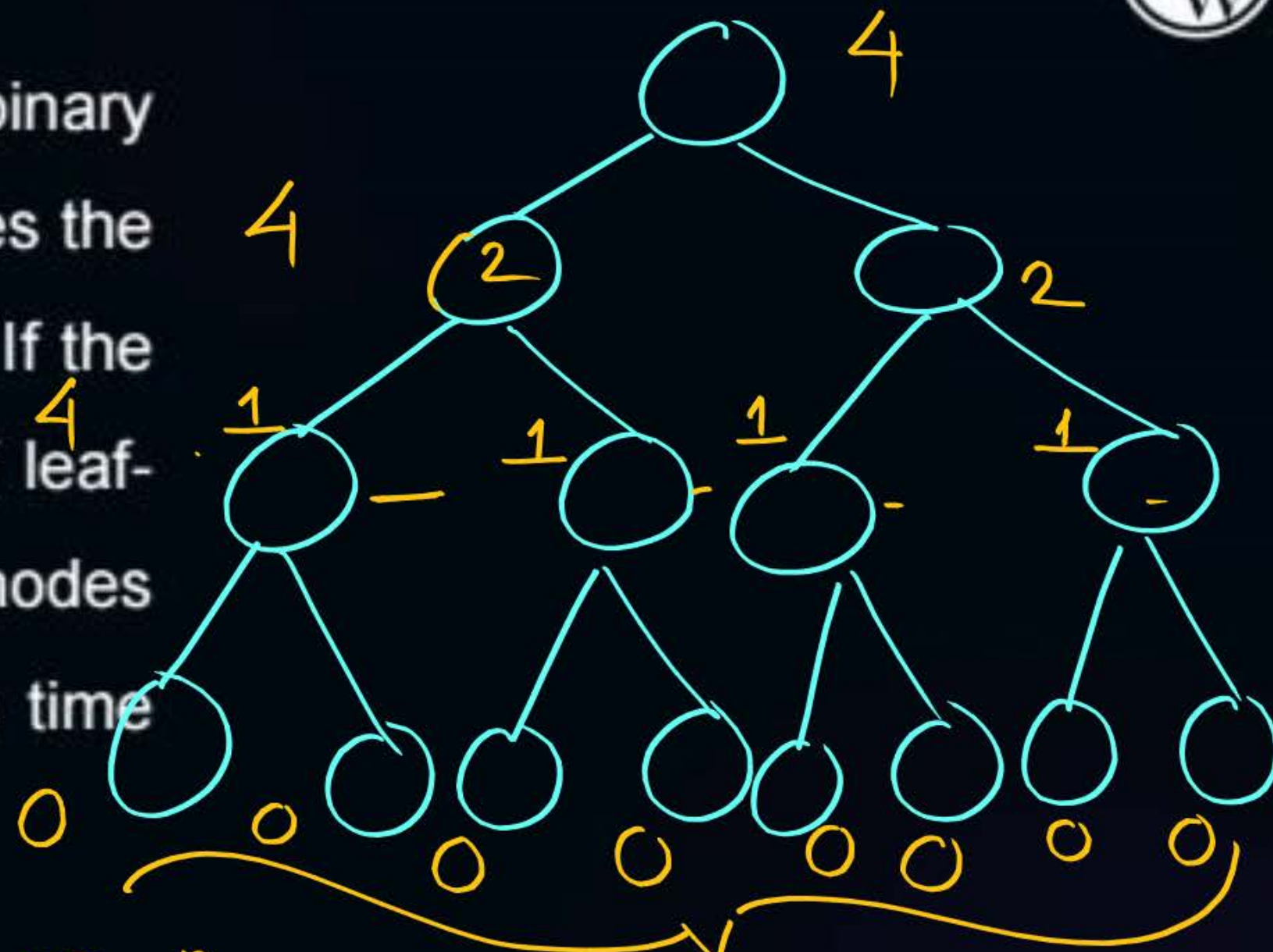
A program takes as input a balanced binary search tree with n leaf nodes and computes the value of a function $g(x)$ for each node x . If the cost of computing $g(x)$ is $\min(\text{number of leaf-nodes in left-subtree of } x, \text{ number of leaf-nodes in right-subtree of } x)$ then the worst-case time complexity of the program is

(A) $\theta(n)$

(B) $\theta(n \log n)$

(C) $\theta(n^2)$

(D) $\theta(n^2 \log n)$



$$\frac{n}{2} \times \log_2 n$$

$$\theta(n \log n)$$

n-Leaf Nodes

$$\frac{n}{2} + \frac{n}{2} + \frac{n}{2} + \dots + \frac{n}{2}$$

#Q For the set of elements {1, 2, 3, 4}: Let X be the number of labelled binary trees that can be formed. Let Y be the number of Binary Search Trees (BSTs) that can be formed. Let W be the number of AVL trees that can be formed. Let Z be the number of binary heaps that can be formed. Find the value of: $X+Y+W+Z$

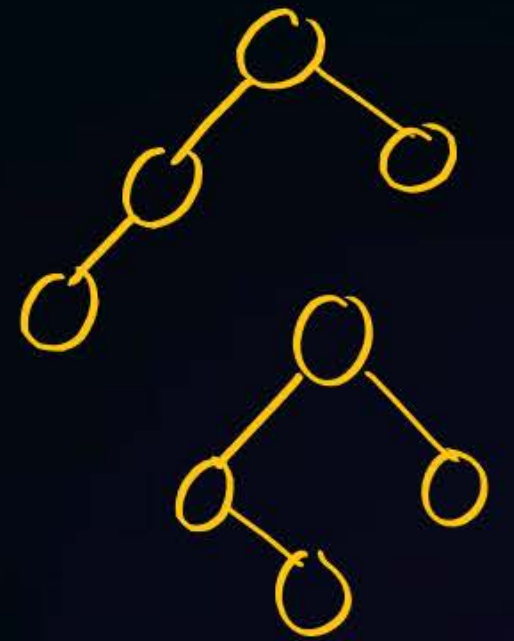
L	R	
3	0	
2	1	— 2
1	2	— 2
0	3	4 ✓
	2	1

$$\text{No. of Labeled trees} = C_4 \times 4! = 14 \times 24 = 336$$

$$\text{No. of BST} = C_4 = 14$$

$$\text{No. of AVL tree} =$$

$$\text{No. of Heaps}$$



#Q For the set of elements {1, 2, 3, 4}: Let X be the number of labelled binary trees that can be formed. Let Y be the number of Binary Search Trees (BSTs) that can be formed. Let W be the number of AVL trees that can be formed. Let Z be the number of binary heaps that can be formed. Find the value of: $X+Y+W+Z$

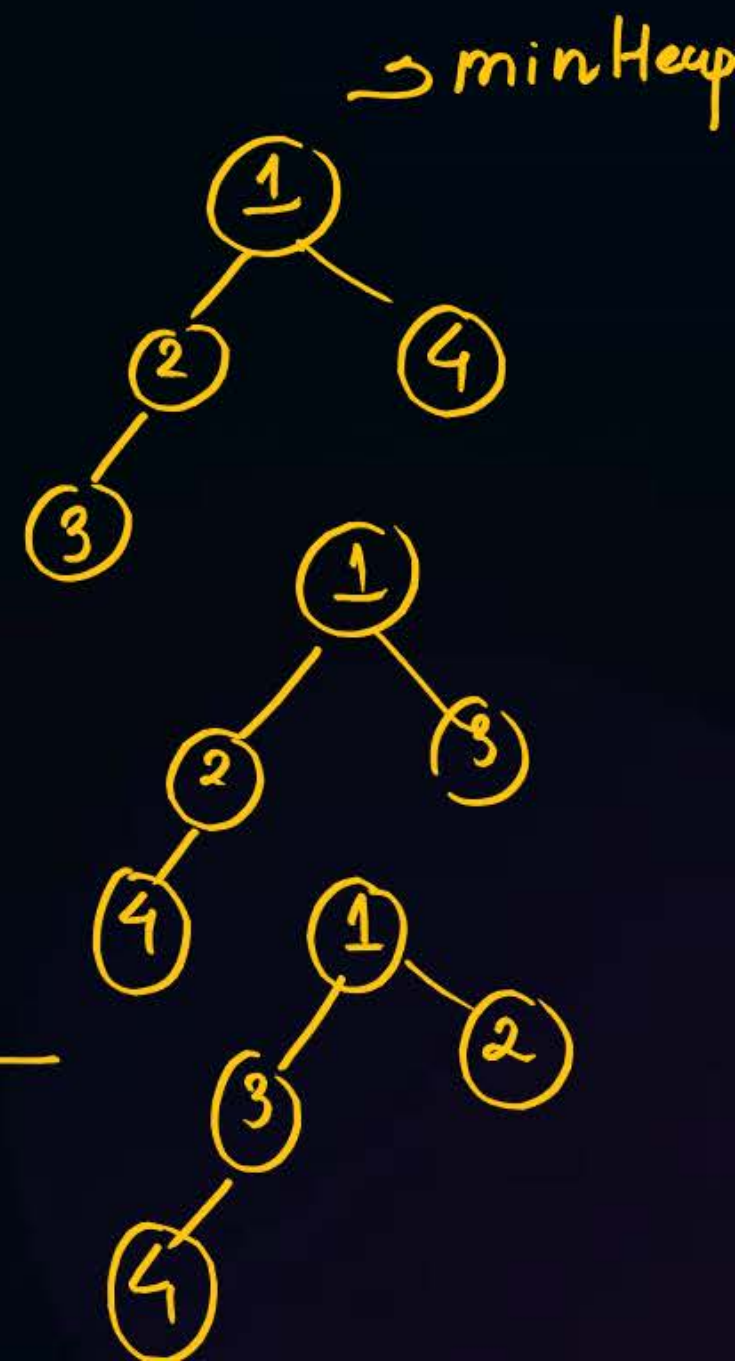
$$\text{No. of Labeled trees} = C_4 \times 4! = 14 \times 24 = 336$$

$$\text{No. of BST} = C_4 = 14$$

$$\text{No. of AVL tree} = 4$$

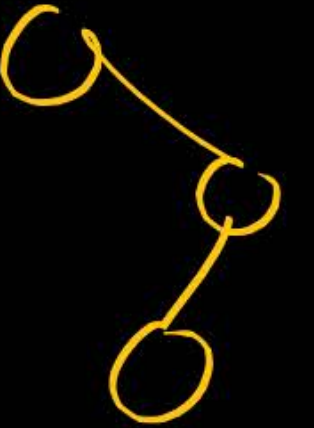
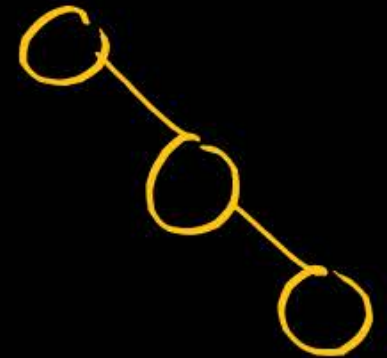
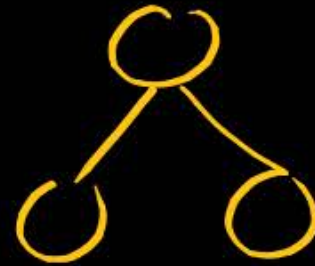
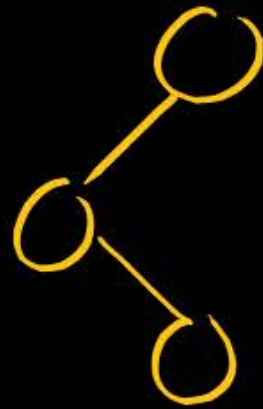
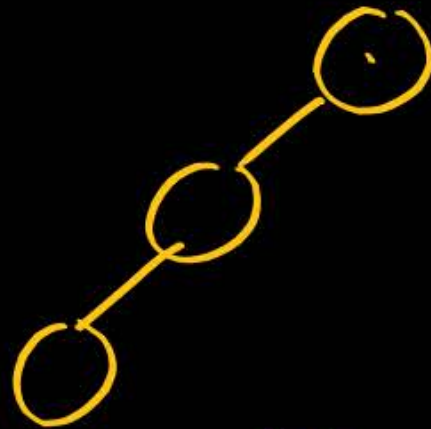
$$\text{No. of Heaps (Min and Max)} = 6$$

$$\underline{360}$$



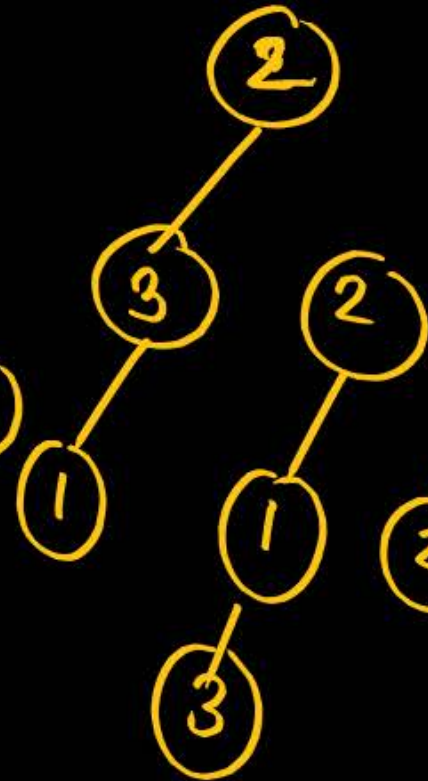
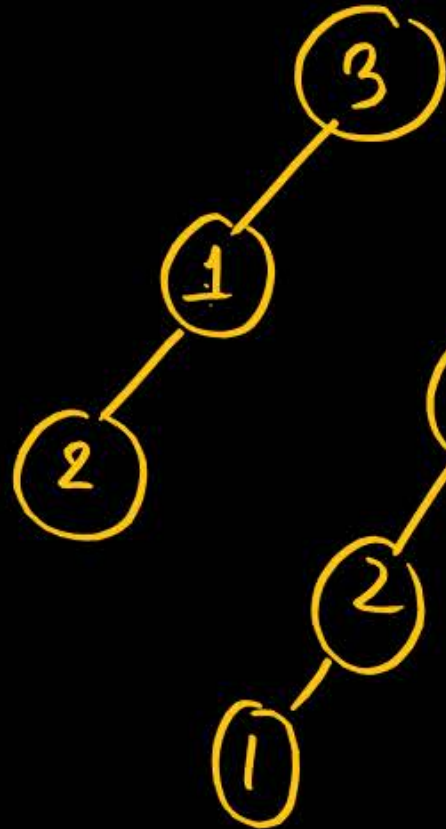
Unlabeled tree is catalan No. C_n

1, 2, 3



Label tree:

$$C_n \times n!$$



5 keys —

Min heap

1. Construct CBT ✓

2. Selection

3. Remove the value

DP

12, 13, 14	11
11, 13, 14	12
11, 12, 14	13
11, 12, 13	14

← 1 key Root (10)

4 keys Left

11, 12, 13, 14

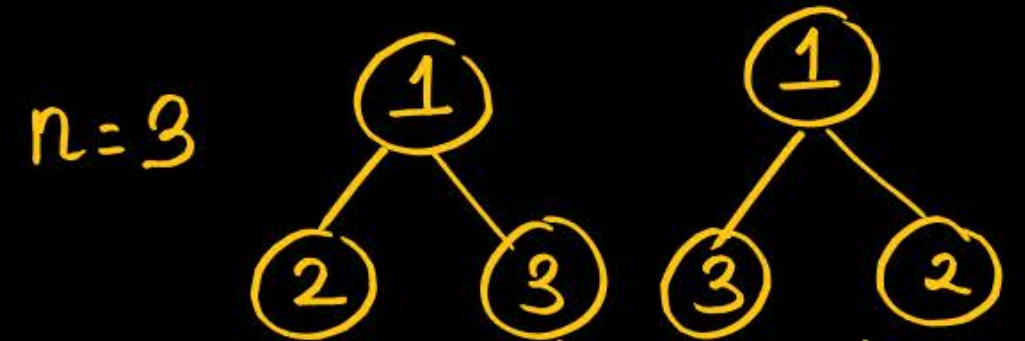
$$4C_3 * 3 \text{ keys} * 1 \text{ key}$$

$$4 \times 2 \times 1 = 8$$

n=1 (1) —

n=2 (1) 1 Heap

(2)



n=4 — 3

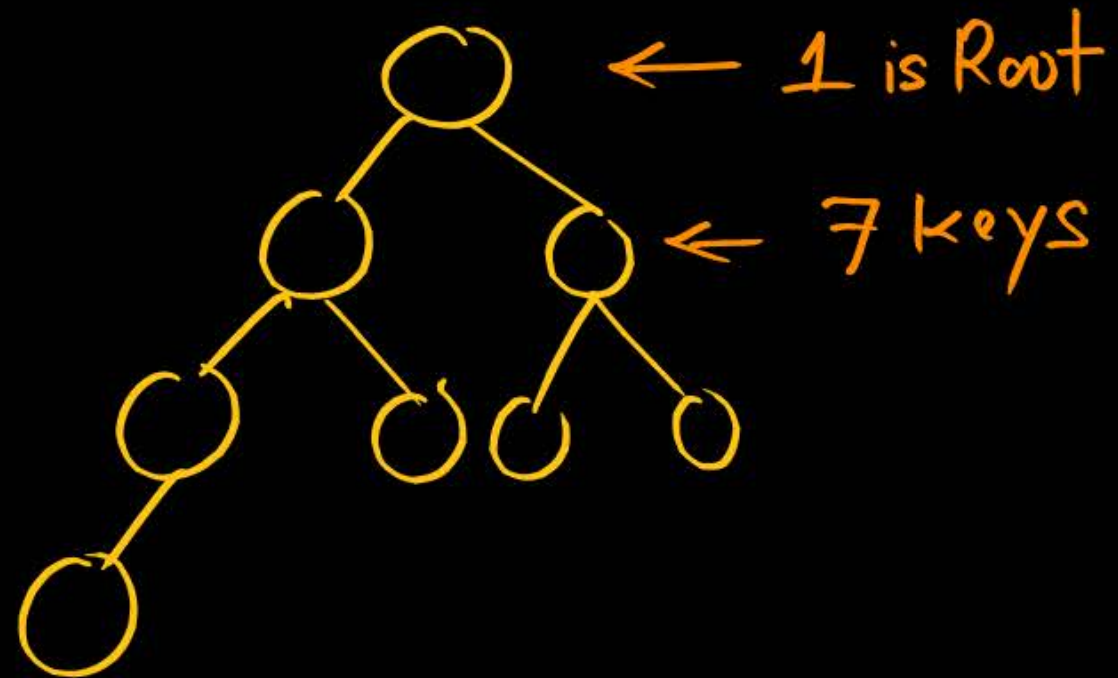
n=5 — 8

n=6 — 20

n=7 — 80

$n=1$	$n=2$	$n=3$	$n=4$	$n=5$	$n=6$	$n=7$
<u>1</u>	<u>1</u>	2	3	8	20	80

No Max heap with 8 Distinct keys ??



$$F_{C_4} = F_{C_3} * \underbrace{n=4}_{\text{Heap}} * \underbrace{n=3}_{\text{Heap}}$$

↓

$$35 \times 3 \times 2 = \textcircled{210}^3$$

Hashing

THANK - YOU